

Unicorn Finance workshop data dictionary

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Authoritative column-level reference for the eight synthetic Unicorn Finance Delta tables in the default namespace `hc_workshop.core_lending`. Types use Databricks SQL notation.

Schema overview

Table	Grain	Purpose	Primary relationship
<code>customer</code>	One borrower or prospect	Customer identity, contact, employment, income, and geography	Parent of <code>loan_application</code>
<code>retail_location</code>	One physical partner store	Merchant and location context for in-store originations	Optional parent of <code>loan_application</code>
<code>loan_product</code>	One product version	Commercial rules, limits, tenors, rates, and interest method	Parent of <code>loan_application</code>
<code>loan_application</code>	One credit request	Origination, promotion, item, associate, and underwriting decision	Child of reference tables; optional parent of <code>credit_contract</code>
<code>credit_contract</code>	One approved application converted to a contract	Accepted financial terms and contract lifecycle	Parent of <code>installment</code> and <code>collection_action</code>
<code>installment</code>	One scheduled fixed-term obligation	Due amounts and servicing state at the configured as-of date	Parent of <code>payment</code> and <code>collection_action</code>
<code>payment</code>	One payment attempt	Posted receipts and failed attempts	Child of <code>credit_contract</code> and <code>installment</code>
<code>collection_action</code>	One action against one delinquent installment	Collection activity, outcome, promise, and attributed payment	Child of <code>credit_contract</code> and <code>installment</code>

Conventions

- **PK** — Primary key.
- **UK** — Unique business key.
- **FK** — Foreign key relationship; Delta does not enforce it automatically.
- **Required** — `Yes` means the generator always supplies a value.
- **Money** — `DECIMAL(18,2)`, denominated in PHP unless stated otherwise.
- **Rates** — `DECIMAL(7,4)` percentage points; `5.9900` means 5.99%.
- **Timestamps** — UTC.
- **PII / sensitive** — Synthetic but suitable for masking, tagging, and access-control exercises.

customer

One row per borrower or prospect.

Column	Type	Required	Key	Description
customer_id	BIGINT	Yes	PK	Surrogate customer identifier
customer_no	STRING	Yes	UK	Human-readable core-system customer number
national_id_no	STRING	Yes	PII	Synthetic government ID
first_name	STRING	Yes	PII	Synthetic given name
last_name	STRING	Yes	PII	Synthetic family name
birth_date	DATE	Yes	PII	Date of birth
sex_code	STRING	No	PII	Source demographic code
mobile_number	STRING	Yes	PII	Synthetic Philippine mobile number
email_address	STRING	No	PII	Synthetic email address
employment_type_code	STRING	Yes		Employment classification
monthly_income_amount	DECIMAL(18,2)	Yes	Sensitive	Declared monthly income
province	STRING	Yes	PII	Current province
city_municipality	STRING	Yes	PII	Current city or municipality

Column	Type	Required	Key	Description
customer_since_date	DATE	Yes		Relationship start date
created_at	TIMESTAMP	Yes		Source creation timestamp
updated_at	TIMESTAMP	Yes		Last source update timestamp

Typical `employment_type_code` values: SALARIED , SELF_EMPLOYED , INFORMAL , UNEMPLOYED , RETIRED .

retail_location

One row per physical store. Merchant fields are denormalized onto the location for the workshop.

Column	Type	Required	Key	Description
store_id	BIGINT	Yes	PK	Store identifier
store_code	STRING	Yes	UK	Source store code
store_name	STRING	Yes		Store display name
merchant_code	STRING	Yes		Parent merchant or chain code
merchant_name	STRING	Yes		Retail partner name
merchant_type_code	STRING	Yes		Merchant classification
city_municipality	STRING	Yes		Store city or municipality
province	STRING	Yes		Store province
region_code	STRING	Yes		Philippine administrative region
opened_date	DATE	Yes		Origination-location start date
active_flag	BOOLEAN	Yes		Whether the location is active
updated_at	TIMESTAMP	Yes		Last source update timestamp

Generated `merchant_type_code` values: LARGE_CHAIN , INDEPENDENT .

loan_product

One row per product version and commercial rule set.

Column	Type	Required	Key	Description
product_id	BIGINT	Yes	PK	Product-version identifier
product_code	STRING	Yes	UK	Product-version business code
product_name	STRING	Yes		Product display name
product_type_code	STRING	Yes		Product family
interest_method_code	STRING	Yes		Interest calculation method
min_principal_amount	DECIMAL(18,2)	Yes		Minimum allowed principal or limit
max_principal_amount	DECIMAL(18,2)	Yes		Maximum allowed principal or limit
min_tenor_months	INT	No		Minimum fixed-term tenor
max_tenor_months	INT	No		Maximum fixed-term tenor
min_monthly_rate_pct	DECIMAL(7,4)	Yes		Minimum monthly percentage
max_monthly_rate_pct	DECIMAL(7,4)	Yes		Maximum monthly percentage
effective_from	DATE	Yes		Product-version start date
effective_to	DATE	No		Product-version end date
active_flag	BOOLEAN	Yes		Whether the product is currently sellable

product_type_code values:

- POS_INSTALLMENT
- CASH_LOAN
- REVOLVING_LINE
- CREDIT_CARD

interest_method_code values: ADD_ON, DECLINING_BALANCE.

Generated product_code values: UNICORN_EASY_4M, UNICORN_POS_A, UNICORN_POS_B, UNICORN_ZERO_PROMO, UNICORN_CASH, UNICORN_CASH_PLUS, UNICORN_FLEX, UNICORN_VISA.

loan_application

One row per credit request and underwriting outcome. Promotion, associate, and financed-item fields are intentionally embedded to keep the workshop at eight tables.

Column	Type	Required	Key	Description
application_id	BIGINT	Yes	PK	Application identifier
application_no	STRING	Yes	UK	Human-readable core-system application number
customer_id	BIGINT	Yes	FK → customer	Applicant
product_id	BIGINT	Yes	FK → loan_product	Requested product version
store_id	BIGINT	No	FK → retail_location	Physical origin; null for digital/direct
sales_associate_id	STRING	No	Sensitive	Synthetic assisting associate number
application_channel_code	STRING	Yes		Origination channel
requested_amount	DECIMAL(18,2)	Yes		Requested principal or credit limit
requested_tenor_months	INT	No		Requested fixed tenor; null for revolving
declared_income_amount	DECIMAL(18,2)	Yes	Sensitive	Income captured at application
underwriting_score	DECIMAL(7,2)	Yes	Sensitive	Point-in-time decision score
decision_code	STRING	Yes		Underwriting outcome
decision_reason_code	STRING	No	Sensitive	Primary decision reason
promotion_code	STRING	No		Embedded promotion identifier
promotion_sponsor_type_code	STRING	No		Promotion funder
item_category_code	STRING	No		POS item category
item_brand_name	STRING	No		POS item brand
financed_amount	DECIMAL(18,2)	No		POS amount financed
submitted_at	TIMESTAMP	Yes		Submission timestamp
decided_at	TIMESTAMP	Yes		Decision timestamp
created_at	TIMESTAMP	Yes		Source creation timestamp
updated_at	TIMESTAMP	Yes		Last source update timestamp

Typical codes:

- application_channel_code: IN_STORE, MOBILE_APP, WEB, CALL_CENTER
- decision_code: APPROVED, DECLINED, CANCELLED

- promotion_sponsor_type_code: BRAND
- item_category_code: SMARTPHONE, APPLIANCE, LAPTOP

credit_contract

One row per accepted contract. Contractual values are retained as snapshots even when they repeat product defaults.

Column	Type	Required	Key	Description
contract_id	BIGINT	Yes	PK	Contract identifier
contract_no	STRING	Yes	UK	Human-readable core-system contract number
application_id	BIGINT	Yes	FK/UK → loan_application	Approved source application
contract_status_code	STRING	Yes		Contract lifecycle status
currency_code	STRING	Yes		ISO currency code; PHP in this dataset
principal_amount	DECIMAL(18,2)	No		Fixed-term financed principal
credit_limit_amount	DECIMAL(18,2)	No		Revolving limit; null for fixed-term
tenor_months	INT	No		Fixed-term tenor
monthly_interest_rate_pct	DECIMAL(7,4)	Yes		Contractual monthly percentage
interest_method_code	STRING	Yes		ADD_ON or DECLINING_BALANCE
processing_fee_amount	DECIMAL(18,2)	Yes		Upfront customer processing fee
subsidy_amount	DECIMAL(18,2)	Yes		Merchant or brand subsidy
origination_date	DATE	Yes		Contract start date
maturity_date	DATE	No		Scheduled maturity for fixed-term products
disbursed_at	TIMESTAMP	Yes		Merchant/customer funding timestamp
closed_at	TIMESTAMP	No		Contract closure timestamp
created_at	TIMESTAMP	Yes		Source creation timestamp
updated_at	TIMESTAMP	Yes		Last source update timestamp

Generated contract_status_code values: ACTIVE, CLOSED, DEFAULTED.

Exactly one of `principal_amount` and `credit_limit_amount` is normally populated:

- Fixed-term POS/cash: `principal_amount`
- Revolving Unicorn Flex/card: `credit_limit_amount`

`installment`

One row per scheduled fixed-term obligation. The pair `(contract_id, installment_no)` is unique.

Column	Type	Required	Key	Description
<code>installment_id</code>	<code>BIGINT</code>	Yes	PK	Installment identifier
<code>contract_id</code>	<code>BIGINT</code>	Yes	FK → <code>credit_contract</code>	Parent contract
<code>installment_no</code>	<code>INT</code>	Yes	UK within contract	Installment sequence
<code>due_date</code>	<code>DATE</code>	Yes		Contractual due date
<code>principal_due_amount</code>	<code>DECIMAL(18,2)</code>	Yes		Principal component
<code>interest_due_amount</code>	<code>DECIMAL(18,2)</code>	Yes		Interest component
<code>fee_due_amount</code>	<code>DECIMAL(18,2)</code>	Yes		Scheduled fee component
<code>total_due_amount</code>	<code>DECIMAL(18,2)</code>	Yes		Total scheduled amount
<code>outstanding_amount</code>	<code>DECIMAL(18,2)</code>	Yes		Current unpaid amount
<code>installment_status_code</code>	<code>STRING</code>	Yes		Servicing status
<code>settled_date</code>	<code>DATE</code>	No		Fully settled date
<code>created_at</code>	<code>TIMESTAMP</code>	Yes		Source creation timestamp
<code>updated_at</code>	<code>TIMESTAMP</code>	Yes		Last source update timestamp

Generated `installment_status_code` values: `SCHEDULED`, `PARTIAL`, `PAID`, `OVERDUE`.

Current DPD is not stored because it changes with the reporting date. Calculate it from `due_date`, `settled_date`, `outstanding_amount`, and a declared as-of date.

`payment`

One row per payment attempt or receipt. Failed attempts remain visible as events.

Column	Type	Required	Key	Description
payment_id	BIGINT	Yes	PK	Payment-event identifier
payment_reference	STRING	Yes	UK	External or source reference
contract_id	BIGINT	Yes	FK → credit_contract	Receiving contract
installment_id	BIGINT	Yes	FK → installment	Applied fixed-term installment
payment_at	TIMESTAMP	Yes		Attempt or receipt timestamp
payment_amount	DECIMAL(18,2)	Yes		Payment amount
payment_channel_code	STRING	Yes		Payment channel
payment_status_code	STRING	Yes		Event status
failure_reason_code	STRING	No		Failure reason when applicable
posted_at	TIMESTAMP	No		Ledger posting timestamp
created_at	TIMESTAMP	Yes		Source creation timestamp

Typical codes:

- payment_channel_code : APP, BANK_TRANSFER, PAYMENT_CENTER, STORE
- payment_status_code : POSTED, FAILED

Only POSTED payments should reduce an outstanding balance. A posted payment should have posted_at ; a failed payment should have failure_reason_code .

collection_action

One row per collection attempt or outcome against a delinquent contract.

Column	Type	Required	Key	Description
collection_action_id	BIGINT	Yes	PK	Collection-action identifier
contract_id	BIGINT	Yes	FK → credit_contract	Contract worked
installment_id	BIGINT	Yes	FK → installment	Specific overdue installment
action_at	TIMESTAMP	Yes		Action timestamp
days_past_due_at_action	INT	Yes	Snapshot	DPD when the action occurred

Column	Type	Required	Key	Description
action_type_code	STRING	Yes		Collection method
outcome_code	STRING	Yes		Action outcome
promise_to_pay_date	DATE	No		Promised payment date
promise_amount	DECIMAL(18,2)	No		Promised amount
amount_collected	DECIMAL(18,2)	Yes		Amount directly attributed to the action
agent_id	STRING	No	Sensitive	Synthetic collection-agent identifier
created_at	TIMESTAMP	Yes		Source creation timestamp

Typical codes:

- action_type_code: SMS, CALL, FIELD_VISIT, NOTICE
- outcome_code: NO_CONTACT, CONTACTED, PROMISE_TO_PAY, PAYMENT_RECEIVED

promise_to_pay_date and promise_amount are populated when outcome_code = 'PROMISE_TO_PAY'. PAYMENT_RECEIVED records reconcile to one POSTED payment with the same installment, timestamp, and amount.

Relationship summary

- loan_application.customer_id → customer.customer_id
- loan_application.store_id → retail_location.store_id
- loan_application.product_id → loan_product.product_id
- credit_contract.application_id → loan_application.application_id
- installment.contract_id → credit_contract.contract_id
- payment.contract_id → credit_contract.contract_id
- payment.installment_id → installment.installment_id
- collection_action.contract_id → credit_contract.contract_id
- collection_action.installment_id → installment.installment_id

Generator metadata

The generator records its version, deterministic seed, as-of date, scale, and shared `workshop.run_id` as Delta table properties rather than adding columns to every business table. Data values are derived from a stable row identifier, field namespace, master seed, and generator version so the same configuration produces the same logical rows across workspaces.